

Proposal for Uncovering the Functional and Ecological Roles of RNA Bacteriophages and Viruses in Soil Ecosystems

1. INTRODUCTION

Soils host some of the most diverse and complex biological systems on Earth, supporting microbial communities that regulate nutrient cycling, carbon storage, and ecosystem stability. While viruses are now recognized as integral components of these systems, most ecological models of soil virology have historically focused on DNA viruses. In contrast, RNA viruses, and RNA bacteriophages in particular, have only recently become accessible to ecological study. Recent syntheses in RNA phage biology emphasize that this historical neglect reflects methodological limitations rather than ecological insignificance, noting that RNA phages represent ancient, diverse, and functionally important components of microbial ecosystems that have only become tractable through meta-omic approaches (Hör et al., 2025; Schackart et al., 2023). This gap has left a major portion of the soil virosphere functionally unexplored, despite growing evidence that RNA viruses are widespread, transcriptionally active, and deeply intertwined with microbial processes (Schackart et al., 2023). The goal of my research is to further investigate RNA bacteriophages in temperate forest soils and to determine how their diversity, persistence, and activity can respond to long-term environmental change, with a specific focus on experimental soil warming.

The central scientific problem motivating my work is that, although RNA bacteriophages are now known to be abundant in soils, their ecological roles remain poorly defined, particularly under sustained climate disturbances. Large-scale metatranscriptomic studies have demonstrated that RNA viruses are present across terrestrial environments and that RNA bacteriophages comprise multiple deeply divergent lineages. Comparative meta-omic analyses further indicate that RNA phages exhibit substantial heterogeneity in genome organization,

replication strategies, and host interaction modes, suggesting that ecological responses to stressors such as warming are unlikely to be uniform across lineages (Hör et al., 2025). However, most existing studies emphasize discovery and taxonomic expansion rather than ecological persistence, stability, or functional consequence. As a result, it remains unclear which RNA bacteriophages consistently associate with soil microbial communities, how they respond to long-term warming, and whether certain viral traits enable persistence under environmental stress. Addressing these questions is critical for understanding how viral communities contribute to microbial regulation and ecosystem function in a changing climate.

My research will focus on RNA bacteriophages identified through RNA-dependent RNA polymerase genes in soil metatranscriptomic datasets from the National Ecological Observatory Network, with data accessed via the Joint Genome Institute's IMG/M platform. The availability of expanded viral reference frameworks, including IMG/VR v4, now allows for environmental RNA viruses to be contextualized using integrated taxonomic, functional, and ecological metadata rather than treated as isolated sequences (Camargo et al., 2024). By leveraging a long-term experimental warming system at Harvard Forest, this work will move beyond snapshot observations and instead examine viral communities under sustained environmental pressure. The primary objective is to compare RNA bacteriophage diversity, abundance, and genome recovery across heated and control plots in order to identify phage populations that persist under warming conditions. Viral operational taxonomic units will be used to cluster RdRP sequences and enable comparative analyses across treatments, while genome-level metrics such as fragmentation and recurrence will serve as proxies for ecological relevance and stability.

The working hypothesis underlying this study is that long-term soil warming acts as a selective force on RNA bacteriophage communities, favoring viral lineages that maintain stable host associations or exhibit functional traits that support persistence

under altered microbial and physicochemical conditions. Recent mechanistic work on RNA phages has demonstrated that some phages encode small regulatory RNAs capable of modulating host replication machinery, providing a plausible molecular basis for differential persistence and host coupling under environmental stress (Silverman et al., 2025). Rather than assuming uniform responses across all RNA phages, this work explicitly tests whether certain RNA bacteriophages remain abundant and transcriptionally active under warming while others decline. By focusing on persistence and recurrence rather than novelty alone, this approach reframes RNA phages as potential contributors to long-term microbial dynamics rather than transient or incidental components of the soil virome.

The significance of my work will lie in its ability to connect recent advances in RNA virus discovery with unresolved ecological questions. Previous research has established that RNA viruses represent a substantial and previously hidden component of environmental viral communities, often referred to as RNA viral dark matter due to their divergence from reference genomes. Recent reviews emphasize that the next phase of RNA phage research must move beyond cataloging diversity and move towards identifying ecological constraints, host specificity, and persistence across environmental gradients (Hör et al., 2025). At the same time, soil viral ecology has increasingly emphasized the need to understand which viruses matter functionally, how they interact with microbial hosts, and how they respond to environmental change. By integrating RdRP-based detection with long-term experimental warming data, this study directly addresses these priorities and extends existing frameworks into a new class of viruses that have been largely overlooked in soil systems.

My work will also build on broader research focused on understanding how climate change reshapes soil microbial and viral communities. Long-term warming experiments at Harvard Forest have already demonstrated that sustained temperature increases can restructure microbial populations, alter carbon cycling, and influence viral assemblages across

multiple viral groups. RNA virus-focused reviews emphasize that RNA phage responses to environmental change are likely mediated through host coupling, replication strategies, and genome-level traits rather than simple presence-absence patterns, reinforcing the need for persistence-focused analysis under sustained perturbation (Hör et al., 2025). By focusing specifically on RNA bacteriophages within this system, the present study adds a new dimension to these findings and complements ongoing work on giant DNA viruses and other viral groups. This continuity strengthens the interpretations of the results and situates RNA phage dynamics within a well-characterized ecological context.

Beyond advancing soil virology, this research will contribute to a general understanding of how viruses mediate microbial responses to climate change. Viruses influence microbial mortality, horizontal gene transfer, and metabolic fluxes, all of which can feed back into ecosystem-scale processes. Identifying RNA bacteriophages that persist under warming conditions provides a foundation for future studies linking viral gene content to functional outcomes, such as nutrient cycling or microbial resilience. In this way, the project bridges molecular-scale viral genetics with ecosystem-level processes, addressing a key challenge in environmental microbiology.

Ultimately, my research seeks to shift the framing of RNA bacteriophages in soils from newly discovered entities to ecologically relevant actors whose persistence and activity can be systematically studied. By combining metatranscriptomics, comparative genomics, and long-term environmental data, the project advances both methodological and conceptual approaches to soil viral ecology. The findings are expected to contribute to a better picture of the soil virosphere and to understand how unseen microbial and viral processes shape ecosystem responses to global environmental change.

2. REVIEW OF LITERATURE OR BIBLIOGRAPHY

Research on environmental viruses has expanded substantially in recent years, yet RNA bacteriophages remain among the least understood components of soil microbial ecosystems. Early work in soil virology is largely centered on DNA viruses, particularly double-stranded DNA bacteriophages, because DNA-based metagenomic approaches dominated the field and provided the most accessible entry point for viral discovery (Williamson et al., 2017; Roux & Emerson, 2022). This historical dependence on DNA-based methods meant that the entire categories of RNA viruses were effectively invisible, not absent, from early ecological models. As a result, foundational models of soil viral ecology emphasized bacterial mortality, nutrient release through lysis, and impacts on carbon cycling driven by DNA phages, while RNA viruses are largely absent from these frameworks. This absence shaped how soil viral function was conceptualized, with the ecological relevance of RNA viruses rarely questioned because it could not be measured. This historical bias has had long-lasting consequences for how soil viral ecology is framed, including which questions are considered mechanistic versus speculative and which viral groups are assumed to contribute meaningfully to ecosystem processes.

This gap began to close with the application of metatranscriptomics and large-scale RNA viral discovery efforts. By sequencing environmental RNA directly, metatranscriptomics enabled detection of RNA viruses through the conserved RNA-dependent RNA polymerase (RdRP) gene, which functions as a universal marker of presence across RNA virus lineages (Neri et al., 2022; Lee et al., 2023). The adoption of RdRP as a detection target represented a conceptual shift from organism-centered to function-centered viral discovery, allowing activity rather than genomic presence to drive interpretation. Global analyses using RdRP-based detection revealed an unexpectedly vast and previously unrecognized RNA virosphere, much of which lacks close representation in reference databases and is often described as RNA viral dark matter (Neri et al., 2022; Lee et al., 2023). These findings demonstrated that RNA viruses are

widespread, diverse, and frequently transcriptionally active in environmental systems, challenging the assumption that DNA viruses dominate soil viral dynamics.

Within global datasets, RNA bacteriophages emerge as a major and ecologically relevant component. The expansion of global RNA virome studies shows that bacteria infecting RNA viruses span multiple deeply divergent clades and are present across terrestrial environments, including soils, at significant abundance (Neri et al., 2022). The use of viral operational taxonomic units to cluster RdRP sequences enabled comparative analyses across samples and treatments, providing a framework for examining patterns of persistence, abundance, and distribution rather than isolated detections. This comparative approach is particularly relevant for environmental perturbation studies, where consistent detection across conditions can be used to infer about some aspects of ecological stability or selective advantage.

Despite major advances in RNA virus discovery, a key limitation of the existing literature is that most environmental RNA virome studies remain largely descriptive, emphasizing diversity and novelty rather than persistence or ecological consequence. Large-scale surveys have convincingly demonstrated that RNA bacteriophages are widespread and transcriptionally active across terrestrial systems, including soils (Neri et al., 2022; Sadiq et al., 2024), yet comparatively few studies have examined how these viruses respond to sustained environmental perturbations. In particular, most work relies on spatial comparisons or short-term sampling, which limits the ability to distinguish transient viral signals from populations that remain stably associated with soil microbial communities. As a result, while RNA bacteriophages are now recognized as abundant components of the soil virosphere, their ecological roles under long-term environmental change remain poorly resolved. This gap is especially relevant in the context of climate warming, where selective pressures may favor viral

lineages with traits that support persistence, efficient host coupling, or resilience to altered microbial dynamics.

As viral RNA virus discovery accelerated, viral taxonomy required substantial revision. The restructuring of the class *Leviviricetes* clarified evolutionary relationships among RNA bacteriophages and established consistent classification criteria based on RdRP phylogeny and associated hallmark genes (Callanan et al., 2021). This work was critical for distinguishing RNA bacteriophages from RNA viruses that infect eukaryotic hosts and for interpreting environmental datasets with confidence. Phylogenetic studies resolving deep evolutionary relationships within the phylum Lenarviricota further demonstrated that RNA bacteriophages exhibit extensive evolutionary diversity and plasticity, suggesting an ability to adapt to heterogeneous and dynamic environments such as soils (Sadiq et al., 2022).

At the same time, soil viral ecology matured into a more integrative discipline that explicitly considers viruses as active components of ecosystem processes. Synthesis of soil virology research emphasized that soil viruses influence microbial diversity, community composition, biogeochemical cycling, and that viral dynamics are sensitive to environmental gradients and disturbance (Williamson et al., 2017; Roux & Emerson, 2022). This framework shifted the field from cataloging diversity toward asking which viruses matter and under what conditions. Empirical studies examining viral communities across soil depth profiles showed that viral populations are stratified along physical and chemical gradients, indicating that persistence and abundance patterns reflect both host availability and environmental constraints (Muscatt et al., 2023).

Climate-focused research further supported the relevance of temperature as a factor in soil viral ecology. Reviews addressing soil viruses in the context of climate change argued that warming has the potential to alter viral-host interactions, viral replication, dynamics, and downstream ecosystem processes (Jansson & Wu, 2022). Experimental studies provided

functional support for this perspective by demonstrating that bacteriophages can influence soil carbon mineralization through effects on bacterial activity, directly linking viral dynamics to carbon cycling (Wang et al., 2022). Together, these findings frame temperature not only as a background variable but as a selective force capable of shaping viral community composition and function.

Long-term experimental warming systems provide a rare and powerful opportunity to move beyond presence-based interpretations and instead evaluate how viral communities reorganize under sustained climate stress. Prior work at Harvard Forest and related warming experiments have demonstrated that prolonged soil warming can potentially restructure microbial communities, alter carbon cycling, and shift viral assemblages across multiple viral groups (Jansson & Wu, 2022; Trubl et al., 2021). Building on this foundation, my work will focus specifically on RNA bacteriophages, using RdRP and genome detection, viral operational taxonomic units, and genome recovery metrics to identify phages that persist across heated and control plots. Rather than treating RNA phages as a homogenous group, this approach allows for the comparison of abundance, recurrence, and fragmentation patterns to infer which viral lineages remain active under warming and which decline. By integrating metatranscriptomic signals with long-term experimental data, this work aims to connect RNA phage diversity with ecological stability, advancing the field from cataloging viruses toward understanding how and why certain RNA bacteriophages can endure more in warmed forest soils.

Metatranscriptomic studies examining viral activity across environmental gradients reinforced the value of RNA-based approaches for ecological inference. As RNA virus detection reflects active transcription rather than relic genetic material, metatranscriptomics allows researchers to focus on viruses that are likely participating in ongoing host interactions (Denison et al., 2011). This distinction is critical in soil systems, where DNA can persist long after biological activity has ceased. Studies comparing viral communities across treatments often

emphasize relative abundance, recurrence across samples, and genome completeness as proxies for ecological relevance and persistence, interpreting consistent detection under stress as evidence of viral resilience or successful host association (Roux & Emerson, 2022; Muscatt et al., 2023).

Advances in computational methods enabled this shift toward large-scale ecological interpretation. Petabase sequence alignment approaches increase sensitivity for detecting rare and divergent RNA viruses across massive datasets (Edgar et al., 2022). Modular viromics pipelines standardized viral identification, clustering, and annotation, improving reproducibility and comparability across studies (Coclet et al., 2024). Error correction and continuing improvement tools addressed the inherent fragmentation challenges of RNA virus assemblies in complex environmental samples, allowing more cautious interpretation of genome completeness and abundance patterns (Song et al., 2025).

Several researchers and research groups have been particularly influential in shaping this field. Global RNA virus discovery efforts led by Neri, Lee, Koonin, and collaborators have established the analytical and conceptual foundations for RdRP-based detection and large-scale comparative analyses. Work on RNA virus taxonomy clarified classification frameworks necessary for ecological interpretation (Callanan et al., 2021). Soil viral ecologists, including Roux, Emerson, Jansson, Wu, and Trubl, integrated viruses into broader ecosystem and climate frameworks, emphasizing functional relevance rather than cataloging alone. Methodological advances by Edgar, Roux, Camargo, and others enabled the scale of discovery required to reveal RNA phage diversity in soils. Their work inspires a shift from asking whether RNA phages exist in soils to asking which ones persist, under what conditions, and why?

These authors have contributed to the literature that motivates a focused investigation of RNA bacteriophages in temperate forest soils under experimental warming. While previous studies established that RNA bacteriophages are diverse, widespread, and active. Relatively

few have examined how these viruses persist across warming treatments or how abundance and genome recovery patterns may reflect ecological resilience. By building on established RdRP-based detection frameworks, vOTU clustering approaches, and soil viral ecology concepts, the present work seeks to extend existing knowledge by focusing on RNA bacteriophage persistence and activity in a well-characterized experimental system. My approach is inspired by the broader shift in soil virology studies linking viral diversity with ecological function and environmental change.

My work is also shaped by the research trajectory of my Principal investigator, Jeffrey Blanchard, who has advanced the study of soil viruses through long-term warming experiments and the discovery of novel giant DNA viruses at Harvard Forest, providing a natural foundation for these questions regarding RNA phages.

3. METHOD/APPROACH/PROCESS [HOW]

This research is being conducted using a computational, genome-resolved metatranscriptomic approach to investigate RNA viruses in soil, with a particular focus on RNA bacteriophages. The project builds on work already completed during earlier stages of the honors sequence (independent studies and 499Y) and extends those analyses toward a more focused examination of viral persistence, abundance, and community structure under long-term environmental warming. All analyses rely on publicly available soil metatranscriptomic datasets generated from a long-term warming experiment at Harvard Forest, accessed through the Joint Genome Institute's IMG/M platform (Roux et al., 2021; Camargo et al., 2023), which provides standardized viral genomic, functional, and ecological metadata for comparative environmental analyses (Sadiq et al., 2024).

To identify RNA viruses, the previously processed soil metatranscriptomic assemblies generated through established NEON and Joint Genome Institute workflows are analyzed.

Assembled contigs are screened for RNA-dependent RNA polymerase genes, which serve as a conserved marker for RNA viruses and enable detection independent of reference genome availability. RdRP-containing contigs meeting length and motif criteria are retained and clustered into viral operational taxonomic units using nucleotide identity thresholds to reduce redundancy and allow comparative analyses across samples, an approach commonly used to organize viral diversity in uncultivated environmental datasets (Andrade-Martinez et al., 2022; Sinno et al., 2025). This is consistent with approaches developed for uncultivated RNA virus populations (Roux & Emerson, 2022; Coclet et al., 2024). Taxonomic assignments are inferred through phylogenetic placement against expanded RNA virus reference trees, with particular attention to bacteriophage-associated lineages, leveraging recent revisions to RNA phage taxonomy and deep evolutionary frameworks (Callanan et al., 2021; Sadiq et al., 2022).

Once viral groups are established, community-level analyses are conducted to compare RNA virus composition across heated and control plots, as well as across soil horizons where relevant. Metrics such as relative abundance, recurrence across samples, and genome fragmentation are used as proxies for viral persistence and ecological relevance rather than simple presence. This idea reflects current recommendations to prioritize stability and activity metrics when interpreting environmental viral communities (Williamson et al., 2017; Roux & Emerson, 2022; Hör et al., 2025). Visualization and exploratory analyses are carried out primarily in R, using tidyverse-based workflows to generate phylum-level summaries and treatment comparisons. Preliminary statistical testing, including contingency-based approaches, has been implemented to assess whether observed differences between treatments exceed random expectations, with analytical decisions informed by prior soil virome studies examining treatment-driven shifts in viral community structure (Muscatt et al., 2023; Trubl et al., 2021).

This work also draws on assembly, read mapping, and normalization strategies developed in prior RNA virus discovery studies and adapted for this dataset, including

approaches designed for fragmented and low-abundance viral genomes in complex environmental samples (Edgar et al., 2022; Coclet et al., 2024). Additional analyses may include deeper examination of viral gene content and putative host associations, including exploratory use of CRISPR spacer matches where feasible, drawing on established computational frameworks for uncultivated phage genome interpretation (Camargo et al., 2023; Roux et al., 2021). Decisions about which analyses are retained will be guided by both biological interpretability and consultation with my faculty sponsor to ensure the scope remains appropriate for an honors thesis.

The primary resources used in this research include high-performance computing (HPC) infrastructure, publicly available metatranscriptomic data from NEON and Harvard Forest, and established bioinformatics tools for sequence alignment, clustering, phylogenetic analysis, and visualization, including databases and workflows developed for large-scale viral genome analysis (Roux et al., 2021)

No human or animal subjects are involved in this research, and no laboratory safety or compliance training is required. My work is completely dry lab-based: specialized training for this project is computational, analytical, and has been obtained through direct mentorship from my principal investigator, prior research experience in microbial and viral genomics (independent studies), and continued independent skill development in R Studio-based data analysis and bioinformatics methods, consistent with training pathways described for computational viromics research (Camargo et al., 2023; Hör et al., 2025). Ongoing feedback from my PI will guide methodological decisions as the project progresses from exploratory analysis toward focused interpretation suitable for the final research manuscript.

4. EVALUATION

The progress and quality of this research will be evaluated through ongoing review by my faculty sponsor and committee members, with emphasis on both research development and the successful completion of a polished Research Manuscript and final presentation. My work will be assessed based on measurable milestones, including the completion of a full literature review, development and application of a reproducible analytical workflow, generation of interpretable results from metatranscriptomic data, and the production of a coherent written manuscript that situates those findings within the broader soil virology literature.

Throughout the semester, my research committee will review progress through bi-weekly meetings, draft submissions, and discussion of interim results. Feedback will be provided verbally during meetings and in written form on outlines, drafts, figures, and analytical decisions. This iterative feedback process will allow my committee to assess whether the scope and direction of the work remain viable and appropriately focused for an Honors Research Manuscript.

The viability of the project will be evaluated based on the ability of the research to produce interpretable results that meaningfully address RNA bacteriophage diversity, persistence, and response to long-term soil warming. Progress toward this goal will be demonstrated through the development of figures, statistical summaries, and comparative analyses suitable for inclusion in both the written manuscript and a final presentation at the Massachusetts Undergraduate Research Conference.

5. COMMUNICATION

Communication with my faculty sponsor will follow a structured and consistent schedule designed to support steady progress and timely feedback. I will meet with my faculty sponsor on a bi-weekly basis for approximately 60 minutes to review analytical progress, discuss results,

troubleshoot challenges, and set short-term goals. These meetings will focus on both technical development and conceptual framing of the research.

In addition to bi-weekly meetings, I participate in weekly lab meetings for a minimum of two hours a week, which provide additional opportunities for discussion, feedback, and exposure to related research projects. Full committee meetings will be scheduled at key points in the semester, particularly during major draft reviews or in preparation for the final presentation.

Between meetings, I am expected to maintain an independent work rhythm consistent with a three-credit Honors Thesis, corresponding to approximately 15+ hours of research per week. My faculty sponsor expects that I arrive at meetings prepared with clear updates, questions, and proposed next steps, and that I actively incorporate feedback into subsequent drafts and analyses.

6. TIMELINE [WHEN]

Progress on this research will be reviewed on a bi-weekly basis through scheduled meetings with my faculty sponsor, with additional feedback provided during weekly lab meetings and draft reviews. This structure is designed to ensure steady progress while allowing flexibility as analyses develop.

The manuscript outline will be completed by February 14, 2026, providing a clear framework for the full Research Manuscript. The first complete draft of the manuscript will be submitted by February 28, 2026, allowing for early substantive feedback on structure, analysis, and interpretation. A second revised draft will be completed by March 31, 2026, incorporating committee feedback and refined analyses.

The final written manuscript will be completed between April 17 and early May 2026, following final revisions and formatting. Research findings will be presented at the

Massachusetts Undergraduate Research Conference in April 2026, which will serve as the primary public presentation of this work. Committee members will be invited to attend this presentation. If needed, an additional informal presentation for committee members can be scheduled near the end of the semester.

7. WORKS CITED

My work will use APA style for in-text citations and bibliography.

- Callanan, J., Stockdale, S. R., Adriaenssens, E. M., Kuhn, J. H., Rumnieks, J., Pallen, M. J., Shkoporov, A. N., Draper, L. A., Ross, R. P., & Hill, C. (2021). Leviviricetes: Expanding and restructuring the taxonomy of bacteria-infecting single-stranded RNA viruses. *Microbial Genomics*, 7(11), 000686. <https://doi.org/10.1099/mgen.0.000686>
- Camargo, A. P., Nayfach, S., Chen, I.-M. A., Palaniappan, K., Ratner, A., Chu, K., Ritter, S. J., Reddy, T. B. K., Mukherjee, S., Schulz, F., Call, L., Neches, R. Y., Woyke, T., Ivanova, N. N., Elie-Fadrosh, E. A., Kyrpides, N. C., & Roux, S. (2023). IMG/VR v4: An expanded database of uncultivated virus genomes within a framework of extensive functional, taxonomic, and ecological metadata. *Nucleic Acids Research*, 51(D1), D733–D743. <https://doi.org/10.1093/nar/gkac1037>
- Andrade-Martínez JS, Camelo Valera LC, Chica Cárdenas LA, Forero-Junco L, López-Leal G, Moreno-Gallego JL, Rangel-Pineros G, Reyes A, 2022. Computational Tools for the Analysis of Uncultivated Phage Genomes. *Microbiol Mol Biol Rev* 86:e00004-21. <https://doi.org/10.1128/mmlbr.00004-21>
- Denison, M. R., Graham, R. L., Donaldson, E. F., Eckerle, L. D., & Baric, R. S. (2011). Coronaviruses: An RNA proofreading machine regulates replication fidelity and diversity. *RNA Biology*, 8(2), 270–279. <https://doi.org/10.4161/rna.8.2.15013>

- Edgar, R. C., Taylor, B., Lin, V., Altman, T., Barbera, P., Meleshko, D., Lohr, D., Novakovsky, G., Buchfink, B., Al-Shayeb, B., Banfield, J. F., de la Peña, M., Korobeynikov, A., Chikhi, R., & Babaian, A. (2022). Petabase-scale sequence alignment catalyses viral discovery. *Nature*, 602(7895), 142–147. <https://doi.org/10.1038/s41586-021-04332-2>
- Hör, J. (2025). Advancing RNA phage biology through meta-omics. *Nucleic Acids Research*, 53(8), gkaf314. <https://doi.org/10.1093/nar/gkaf314>
- Jansson, J. K., & Wu, R. (2023). Soil viral diversity, ecology and climate change. *Nature Reviews. Microbiology*, 21(5), 296–311. <https://doi.org/10.1038/s41579-022-00811-z>
- Limited effect of short- to mid-term storage conditions on an Australian farmland soil RNA virome. (n.d.). <https://doi.org/10.1128/jvi.01459-25>
- Lee, B. D., Neri, U., Roux, S., Wolf, Y. I., Camargo, A. P., Krupovic, M., RNA Virus Discovery Consortium, Simmonds, P., Kyrpides, N., Gophna, U., Dolja, V. V., & Koonin, E. V. (2023). Mining metatranscriptomes reveals a vast world of viroid-like circular RNAs. *Cell*, 186(3), 646-661.e4. <https://doi.org/10.1016/j.cell.2022.12.039>
- Muscatt, G., Cook, R., Millard, A., Bending, G. D., & Jameson, E. (2023). Viral metagenomics reveals diverse virus-host interactions throughout the soil depth profile. *mBio*, 14(6), e0224623. <https://doi.org/10.1128/mbio.02246-23>
- Neri, U., Wolf, Y. I., Roux, S., Camargo, A. P., Lee, B., Kazlauskas, D., Chen, I. M., Ivanova, N., Zeigler Allen, L., Paez-Espino, D., Bryant, D. A., Bhaya, D., RNA Virus Discovery Consortium, Krupovic, M., Dolja, V. V., Kyrpides, N. C., Koonin, E. V., & Gophna, U. (2022). Expansion of the global RNA virome reveals diverse clades of bacteriophages. *Cell*, 185(21), 4023-4037.e18. <https://doi.org/10.1016/j.cell.2022.08.023>
- Roux, S., & Emerson, J. B. (2022). Diversity in the soil virosphere: To infinity and beyond? *Trends in Microbiology*, 30(11), 1025–1035. <https://doi.org/10.1016/j.tim.2022.05.003>

- Sadiq, S., Chen, Y.-M., Zhang, Y.-Z., & Holmes, E. C. (2022). Resolving deep evolutionary relationships within the RNA virus phylum Lenarviricota. *Virus Evolution*, 8(1), veac055. <https://doi.org/10.1093/ve/veac055>
- Sadiq, S., Harvey, E., Mifsud, J. C. O., Minasny, B., McBratney, A. B., Pozza, L. E., Mahar, J. E., & Holmes, E. C. (2024). Australian terrestrial environments harbour extensive RNA virus diversity. *Virology*, 593, 110007. <https://doi.org/10.1016/j.virol.2024.110007>
- Schackart, K. E., Graham, J. B., Ponsero, A. J., & Hurwitz, B. L. (2023). Evaluation of computational phage detection tools for metagenomic datasets. *Frontiers in Microbiology*, 14. <https://doi.org/10.3389/fmicb.2023.1078760>
- Silverman, A., Nashef, R., Wasserman, R., Noy, T., Born, S., Yao, T., Geng, Y., Rotbard, H., Levkowitz, A., Kaufman, Y., Golding, I., & Melamed, S. (2025). Phage-encoded small RNA hijacks host replication machinery to support the phage lytic cycle. *Molecular Cell*, 85(24), 4678-4697.e12. <https://doi.org/10.1016/j.molcel.2025.11.019>
- Sinno, A., Baghdadi, R., Narch, R., El Rayes, S., Tokajian, S., & Al Khoury, C. (2025). Charting the virosphere: Computational synergies of AI and bioinformatics in viral discovery and evolution. *Journal of Virology*, 99(12), e0155425. <https://doi.org/10.1128/jvi.01554-25>
- Song, H., Tithi, S. S., Brown, C., Aylward, F. O., Jensen, R., & Zhang, L. (2025). Virseqimprover: An integrated pipeline for viral contig error correction, extension, and annotation. *PeerJ*, 13, e18515. <https://doi.org/10.7717/peerj.18515>
- Trubl, G., Kimbrel, J. A., Lique-Gonzalez, J., Nuccio, E. E., Weber, P. K., Pett-Ridge, J., Jansson, J. K., Waldrop, M. P., & Blazewicz, S. J. (2021). Active virus-host interactions at sub-freezing temperatures in Arctic peat soil. *Microbiome*, 9(1), 208. <https://doi.org/10.1186/s40168-021-01154-2>

Williamson, K. E., Fuhrmann, J. J., Wommack, K. E., & Radosevich, M. (2017). Viruses in Soil Ecosystems: An Unknown Quantity Within an Unexplored Territory. *Annual Review of Virology*, 4(1), 201–219. <https://doi.org/10.1146/annurev-virology-101416-041639>